Computer Organization and Architecture

Assignment - 1

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Q1. What is Raspberry Pi?

Answer:

Raspberry Pi is a mini, cheap computer produced by the Raspberry Pi Foundation. Although it is the size of a credit card, it can do most of the simple things that a normal computer can do. Primarily used to learn programming, electronics, robotics, IoT and AI projects. Raspberry Pi has the ability to run in different operating systems, and Raspberry Pi OS is the most popular. It is a very inexpensive, portable, and low power consuming device, which is widely used by students, hobbyists and developers to create innovative projects and experiment with new technologies.

Q2. So, what were the top 10 learnings from the AI Summit?

Answer:

Some important takeaways from the AI Summit are:

- Technology is rapidly evolving with AI in all areas.
- Good data quality is a direct factor in the performance of AI.
- AI is bringing transformation in healthcare and education.
- AI is helping to make cybersecurity smarter.
- AI is generating new jobs.AI is inventing new careers.
- Good computing facilities are essential.
- Human skills and Artificial Intelligence should go hand-in-hand.

- There's a need for ongoing learning because AI is changing rapidly.
- This is just another indication of how AI will continue to grow in significance in the future.

Q3. What is YOTTA?

Answer:

Yotta is an Indian company offering cloud computing and data centre services. It secures, stores, and processes data for various businesses. The company also provides services like cloud hosting, disaster recovery, cybersecurity, and GPU-based computing for Artificial Intelligence applications. Yotta is used by many organisations due to its dependability and scalability, without requiring costly hardware investments. Yotta is poised to be a key contributor to cloud computing, artificial intelligence advancements, and the overall functioning of businesses in India as the country embraces digital transformation.

Q4. What is NPTEL?

Answer:

NPTEL is the National Programme on Technology Enhanced Learning. It's an online learning platform initiated by the IITs and IISc, and supported by the Government of India. It provides free Engineering, Science, Management and many other courses. Students will have the opportunity for self-paced video lectures, assignments, and quizzes. A certificate is awarded to the successful candidates of the final examination. Engineering students have been using NPTEL for bettering their technical knowledge, learning new skills and preparing for internships and placements.

Q5. What are the criteria to choose a programming language to create an application? What are the parameter(s) of the system?

Answer:

The programming language used depends on the application that is being created. In AI, Python is used, while in enterprise applications, Java is the preferred choice. JavaScript for web development and C++ for system programming. When choosing a language, developers take into account its performance, security, coding ease,

platform support, scalability, and community support. Libraries and future maintenance also are important. Choosing the proper language is critical when it comes to creating an application which is efficient, reliable, and simple to expand as needs evolve.

Q6. What does RAM Size and Word Size mean? What is the speed and operating system of the system if it is 64 bit.

Answer:

The amount of temporary memory present in the computer to run programs is called RAM. Typically reported in gigabytes (GB). The more RAM you have, the more multitasking and smooth performance you'll have. The size of a word is the number of bits that a processor can process in a single operation. The processor in a 64-bit system processes 64 bits of data at a time and is more efficient than a 32-bit system because of this. The processor speed is measured in GHz (gigabytes per second) and common 64-bit operating systems include Windows, Linux and macOS.

Q7. Why is GPU quicker than CPU?

Answer:

A CPU can work on other tasks in sequence, whereas a GPU can work on numerous tasks at once. This is because GPUs have hundreds or thousands of smaller cores that are used in parallel, while a CPU has a few powerful cores. This parallel processing speed makes GPUs particularly fast in graphics rendering, video editing and AI applications. But CPUs remain vital to the functioning of the operating system and to the effective performance of the computer for day-to-day use.

Q8. Explain what each of the following means: CPU, GPU and TPU.

Answer:

The brain of a computer, the CPU or Central Processing Unit, is responsible for running the computer's instructions and managing all operations. GPU (Graphics Processing Unit): A specially designed processor for graphics and parallel computing, which is beneficial for gaming and artificial intelligence. The TPU (Tensor Processing Unit) is a special-purpose processor designed by Google to handle tasks related to

Artificial Intelligence and Machine Learning. In AI applications, it can handle complex mathematical calculations much quicker and more efficiently than a CPU or GPU.

Q9. What is Turing Award and who was awarded the Turing Award last year?

Answer:

The Turing Award is the highest honour in the field of Computer Science. It is presented every year by the Association for Computing Machinery (ACM) to people who have made outstanding contributions to computing and technology. Sometimes referred to as the "Nobel Prize of Computing. The 2025 Turing Award is shared by Andrew G. Barto and Richard S. Sutton, for their groundbreaking contributions to Reinforcement Learning, a field that has significantly advanced the development of today's AI.

Q10. What is the Stampeding Herd Problem?

Answer:

The Stampeding Herd Problem is the case in which many users, processes or threads access the same resource simultaneously. This typically occurs when a cached object expires, or a blocked resource becomes available. Consequently, a bunch of requests are delivered to the server all at once, leading to high system load and poor performance. To prevent this, techniques such as caching, locking, queuing and load balancing are implemented. These techniques are useful for efficient distribution and performance enhancement of requests.

Q11. What type of CPU configuration does your laptop computer have?

Answer:

The computer I am using is the Von Neuman type, the most popular type of CPU structure in modern computers. In this architecture, instructions and data occupy the same memory and are executed sequentially by CPU. The CPU is composed of the Control Unit (CU), the Arithmetic Logic Unit (ALU), and the Registers that help in the processing of instructions efficiently. This architecture is simple, reliable and good for general purpose computing, making it popular in most personal computers and laptops.

Q12. What organization did Intel and AMD founder?

Answer:

In 1968, Intel was founded by Robert Noyce and Gordon Moore. It emerged as one of the world's premier computer processor and semiconductor device manufacturers. In 1969, AMD (Advanced Micro Devices) was established by Jerry Sanders and his 7 co-founders. However, today Intel has one of its most significant rivals in the form of AMD, and is familiar for its Ryzen and EPYC processors. Both companies have played an important role in the evolution of modern computer technology and high-performance processors.

Q13. Imitation Game – What did you learn after watching this movie?

Answer:

Following this movie, The Imitation Game, I learned that if you are determined, you are patient, and have logical thinking, you can solve a difficult problem. Alan Turing was confronted with many challenges during World War II, but he did not stop and he did not give up but always worked towards his goal. He helped to crack the Enigma code, which revolutionized history and helped to develop modern computer science. Another lesson I learned from the film is to trust in your ideas even if others don't believe in you; to collaborate with others to achieve your goals; and to be inventive. It was an inspirational and motivational experience.